
UNIVERSITY OF CAPE TOWN

Test - 28 July 2018

Portfolio Optimization: ECO 4102Z

2 hours 30 minutes

Maximum Marks: 75

INSTRUCTIONS TO CANDIDATES

- i. Answer ALL questions
 - ii. Write legibly, using blue or black ink
 - iii. Show all your workings
 - iv. You may bring own formula sheet
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1. During the past five years, you owned two stocks that had the following annual rates of return:

Year	Stock A	Stock B
1	0.19	0.08
2	0.08	0.03
3	-0.12	-0.09
4	0.03	0.02
5	0.15	0.04

- (a) Compute the arithmetic mean annual rate of return for each stock. Which stock is most desirable by this measure? [3 marks]
- (b) Compute the standard deviation of the annual rate of return for each stock. By this measure, which is the preferable stock? [5 marks]
- (c) Compute the coefficient of variation for each stock. By this relative measure of risk, which stock is preferable? [3 marks]
- (d) Under what conditions can the standard deviation be used to measure the relative risk of two investments? [1 marks]

- (e) Under what conditions must the coefficient of variation be used to measure the relative risk of two investments? [1 marks]

[Total: 13 marks]

2. Suppose an investor measures the utility function of her wealth (w) using the exponential utility function $U(w) = -e^{-\beta w}$. For $w > 0$, and $\beta > 0$:

- (a) Assess whether the function has a decreasing, increasing or constant
- Absolute risk aversion
 - Relative risk aversion

[3 marks]

- (b) Based on your answer in (a), above, comment on what this tells us about,

- the amount and
- the proportion

of her assets that this investor will invest in risky assets.

[2;2 marks]

[Total: 7 marks]

3. (a) Explain the type of preferences that an investor should have in order to use: *i.*) First-order stochastic dominance criterion, *ii.*) Second-order stochastic dominance criterion, in making an investment choice decision. [1, 2 marks]

Consider the following investments:

Investment 1		Investment 2	
Payoff	Probability	Payoff	Probability
R1m	0.25	R3m	0.33
R7m	0.50	R5m	0.33
R12m	0.25	R8m	0.34

- (b) If preferences are based on the mean and variance, which investment would an investor choose? [4 marks]
- (c) According to the second-order stochastic dominance criterion, how would you compare the two investments? [5 marks]

[Total 12 marks]

4. An investor wishes to allocate her capital between a service company S and a manufacturing company M . The investor believes that returns on shares in S have mean 10% and variance 16% while returns on shares in M have mean 8% and variance 25%. The correlation between the two companies is 0.3. Assume the investor chooses their investments according to mean-variance portfolio theory.

- (a) Explain which company's share she would prefer. [2 marks]

Assume the investors preferences are described by a standard quadratic utility function.

- (b) State which assumption of the mean-variance portfolio theory can be relaxed. [1 mark]

- (c) Calculate the expected return and the standard deviation of a portfolio which is invested three quarters in S and one quarter in M . [2 marks]

- (d) Calculate the minimum variance portfolio. [2 marks]

A new study suggests that in the future, S will make more sales to M , when M is delivering strong profits.

- (e) Describe the effect this will have on portfolios composed of M and S , including the minimum-variance portfolio. [3 marks]

[Total: 10 marks]

5. (a) In the context of portfolio theory, define the following an "efficient portfolio" portfolio. [1 mark]
- (b) Assume analysts provide the following types of information (see Table 1 below). Assume short sales are allowed (standard definition). Determine the expected return and standard deviation of the optimum portfolio if the lending and borrowing rate is 5%. [10 marks]

Table 1: Use this Information for part (c)					
Stock	Mean	Standard	Covariance with		
	Return(%)	Deviation(%)	1	2	3
1	10	4	20		40
2	12	10			70
3	18	14			

[Total 11 marks]

6. In a market where the assumptions of the Capital Asset Pricing Model (CAPM) hold, there are a risk-free asset and two risky assets with the following attributes:

State	Probability	Rate of return (per annum)		
		Asset 1	Asset 2	Asset 3
1	0.2	5.0%	15.0%	26.0%
2	0.3	5.0%	22.0%	15.0%
3	0.1	5.0%	10.0%	24.0%
4	0.4	5.0%	28.0%	7.0%
Market capitalisation			R30,000,000	R70,000,000

- (a) Determine the composition of the market portfolio. [1 mark]
- (b) Determine the market price of risk. [5 marks]
- (c) Calculate the beta of each risky asset. [2 marks]
- (d) Using your calculations above (or otherwise) comment on whether each of the risky assets is under or over-performing. As an investor what position would you take on each of the risky assets? [2;1 marks]
- [Total 11 marks]

7. You are the lead manager of a large mutual fund. You have become aware that several equity analysts who have recently joined your management team are interested in understanding the differences between the capital asset pricing model (CAPM) and the Multi-index Model. In particular, they are interested in how these two asset pricing models can help them perform better security analysis.

- (a) Explain how a multi-index model would be expected to perform relative to a single-index model in terms of fitting data and predicting future security price movements. [2 marks]

Consider the following two-factor model for the returns on three assets A, B and C :

Asset	A	B	C
α_i	0.03	0.05	0.1
$b_{i,1}$	1	3	1.5
$b_{i,2}$	-4	2	1.5

- (b) Determine the equation for the expected return on a portfolio which has weights (x) , $x_A = -0.5, x_B = 1.5, x_C = 0$. [4 marks]
- (c) Construct a portfolio of securities A, B, C that has a factor beta of 1 on the first factor and 0 on the second factor, i.e. the expected return on the portfolio is:

$$R_P = \alpha_P + I_1 + \epsilon_i$$

Comment about the characteristics of this portfolio [3,2 marks]

[Total: 11 marks]